SRT-Net: Scattering Region Topology Network for Oriented Ship Detection in Large-Scale SAR Images

Dece Pan, Xin Gao, Wei Dai, Jiamei Fu, Zhirui Wang, Xian Sun, *Senior Member, IEEE*, and Youming Wu

***Abstract*— Synthetic aperture radar (SAR) ship detection plays an important role in the field of maritime security. However, certain unique imaging properties make it challenging to extract the shape features of ships, such as speckle noise and strong scattering interference from irrelevant objects. These factors result in inaccurate ship localization and obvious false alarms under complex large-scale inshore scenes. To address this issue, we propose the scattering region topology network (SRT-Net), which can dynamically capture the comprehensive global context and enhance the ship saliency. This is achieved through two key modules, namely the scattering region topological structure pyramid (SRTP) and the ship saliency enhancement (SSE) module. The former provides richer semantic information to distinguish the object from the background, while the latter offers an extra pixel-level classification task to guide accurate bounding box regression. Thanks to the guidance of richer information, the proposed method can achieve fewer false alarms and enhance location accuracy. Additionally, we introduce a scale feature adaptive (SFA) loss to balance the attention to ships with various scales, which improves the robustness of multiscale ship detection. The proposed method achieves state-of-the-art performance under complex inshore scenes, and its effectiveness is verified by experiments on a large-scale SAR ship detection dataset (LSSDD) and a public SAR ship detection dataset (SSDD+).**

***Index Terms*— Deep learning, graph convolution network, oriented ship detection, strong scattering interference, synthetic aperture radar (SAR), topological structure.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an advanced microwave sensor that works in all weather conditions and is available all day long. It has become a critical tool in the maritime surveillance domain, such as marine traffic control, fisheries management, and shipwreck rescue [1], [2]. Ship detection is crucial for maritime surveillance as the activities of ships directly impact international trade and maritime security. Recently, with the launch of a series of SAR satellites, a large amount of data has driven SAR ship detection to be a hotspot.

The traditional SAR ship detection method typically involves four stages: preprocessing, sea–land segmentation, candidate region extraction, and object discrimination [3]. In the first two stages, some algorithms are used to filter out imaging noise and separate the sea from the land to highlight the ship targets and reduce the interference from the land area [4], [5]. The candidate regions extraction stage applies ship detection methods to identify areas in the SAR image that may contain ships. Then, the object discrimination stage filters out ship objects from the candidate areas to reduce false alarms. The constant false alarm rate (CFAR) [6] is a widely used discrimination method that models the background clutter statistically and filters the pixels of the candidate region to extract ship objects. Based on CFAR, several improved methods have been proposed. Ai et al. [7] introduce a novel method for modeling the grayscale distribution of ocean clutter to reduce false alarms. Mamgain et al. [8] combine coarse-stage and fine-stage CFAR to avoid missed detection due to target occlusion. Hou et al. [9] present a multilayer CFAR with log-normal distribution to increase positioning accuracy and computation efficiency. Even though traditional methods have achieved reasonable performance in some simple scenes, they lack generalizability to complex scenes and heavily rely on manual data analysis.

Deep learning methods with automatic feature extraction have gained popularity in the SAR community. Unlike traditional methods, deep learning methods can extract features from SAR images without preprocessing and sea–land segmentation stages. They also have exceptional generalization ability and simplicity, making them attractive for SAR applications. Most mainstream deep learning methods are designed for natural scene applications [10], [11], so they mainly use horizontal bounding boxes (HBBs) representation to indicate the precise position, as shown in Fig. 1(a). However, the HBB representation is not suitable for long and narrow objects with a large aspect ratio, such as ships. It lacks the ability to convey the angle and size information of ships directly. Moreover, it tends to contain redundant background information within the bounding box, making it challenging for the model to extract ship features accurately during training. Consequently, the model may mistakenly identify artificial facilities or other objects as ships, leading to a high number of false alarms.

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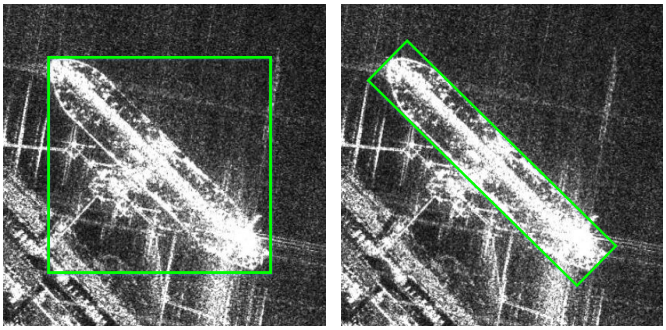
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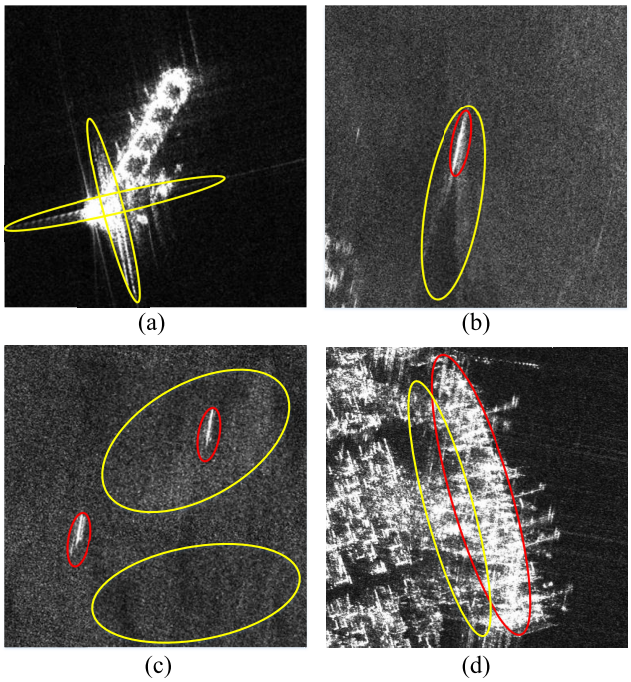
(a) (b)

Fig. 1. (a) Horizontal bounding box. (b) OBB.

To address the aforementioned limitations, several SAR ship detection methods employing oriented bounding box (OBB) have been proposed [12], [13], [14], which provide more refined localization results, as shown in Fig. 1(b). However, these methods still face some challenges that hinder their effectiveness in SAR ship detection.

First, these methods expand the receptive field by stacking convolution and pooling operations, but the receptive field is still limited. When the ship target and the surrounding irrelevant objects have similar visual characteristics and scattering mechanisms, it can be challenging to determine whether a strong scattering region (SSR) belongs to the ship or an irrelevant object without semantic information outside the local region. Moreover, the strong scattering interference from irrelevant objects also poses a challenge for the detector to extract the contour features of ships, potentially resulting in inaccurate localization or missed detection. Second, the imaging noise and the strong scattering interference, as shown in Fig. 2, may obscure the contour information of ships. And they may cover the low-scattering parts of the ship, such as the deck, dramatically increasing the difficulty of predicting the size and contour information of ships. Third, these methods lack an adaptive balance learning strategy for multiscale ships, which makes the detector more inclined to learn the features of large ships. This phenomenon results in unbalanced learning of size features. Furthermore, anchor-based ship detection methods require dense sampling to generate multiple anchors. However, ships are sparsely distributed in SAR images, so most of the anchors are redundant, resulting in a considerable computational burden.

Based on the above analysis, a scattering region topology network (SRT-Net) is proposed to detect arbitrarily oriented SAR ships in large-scale and complex scenes. Unlike previous methods that directly apply deep learning methods to SAR ship detection, the SRT-Net identifies SSRs and builds topological structures for them. Concretely, it uses graph convolutional networks (GCNs) to transfer feature similarity across key regions and extract global context from the entire SAR images in an adaptive manner. The extracted global context can reflect the structural differences across SSRs and provide richer semantic information, enhancing the separability of ships and background and suppressing false alarms. Besides, the SRT-Net utilizes a ship saliency enhancement module (SSE) to extract ship contour information and learn geometric information from ships to highlight the saliency of ships in complex scenes. To mitigate the issues of imbalanced



(a) (b) (c) (d)

Fig. 2. Some typical interference of ship detection in SAR images. The yellow and red oval represent the interference and ship objects, respectively. (a) Sidelobes. (b) Wakes. (c) Ocean clutter. (d) Strong scattering interference from artificial facilities.

scale feature learning, a scale feature adaptive (SFA) loss is introduced. This loss function considers the size information of ships and increases the attention on small-scale ships. Moreover, the SRT-Net uses an anchor-free frame with a set of vectors to indicate the OBB, significantly reducing the number of hyperparameters required.

Our main contributions are as follows.

1) The scattering region topological structure pyramid (SRTP) constructs a topological structure for SSRs, allowing a broader receptive field to extract global context from the entire SAR image adaptively. It uses GCNs to extract feature similarities across regions, thus exploiting structural differences between ships and other objects to improve the separability of ships and background and reduce false alarms.
2) To enhance the saliency of ships in various imaging conditions, an SSE module is proposed which adds an additional pixel-level classification task of ship contours. The designed task guides the model in learning the geometric information of ships and further improves the localization accuracy.
3) The SFA loss is introduced to balance the attention to ships of different scales by fusing the object scale information. It enhances the focus on small-scale ships that are easily missed, effectively improving the robustness of multiscale ship detection.

To validate the effectiveness of SRT-Net, extensive experiments are carried out, and our method achieves satisfactory results compared to current competitive methods. The following is the organization of this article. In Section II, a review of related work is provided. In Section III, detailed descriptions of our work are presented. In Section IV, the large-scale SAR ship detection dataset (LSSDD) and the public SAR ship

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detection dataset (SSDD+) are introduced, and the results and analysis of the related experiments are provided. In Section V, the conclusion is drawn.

II. Related Work

A. Deep Learning-Based Object Detectors

In the area of object detection, deep convolutional neural networks (DCNN) have achieved landmark success due to their ability to extract rich features. The current deep learning-based object detectors can be classified into two types based on their detection heads: two-stage detectors and single-stage detectors.

The two-stage detectors typically start by obtaining the proposal box from the original image, then classify the proposal box and refine it to produce accurate bounding boxes. Faster R-CNN [11] is a representative of two-stage detectors, which designs a region proposal network (RPN) to generate candidate regions directly, making the whole process trainable end-to-end. To improve detection efficiency, R-FCN [15] adopts an efficient fully convolutional network structure to avoid complex independent computation for each proposal region. Later, cascade R-CNN [16] introduces a cascaded network structure and sets different thresholds for each network, enhancing the overall detection accuracy.

Several single-stage detectors have gained attention for their computational efficiency, including YOLOv3 [17], SSD [18], and RetinaNet [19]. These methods simplify the whole pipeline into a single stage, eliminating the need to generate candidate regions and refinements, leading to faster computation. However, these single-stage network structures have lower accuracy than two-stage detectors. All the methods above are anchor-based detectors that need to generate many anchors and set their parameters manually, which significantly impacts the detection performance and efficiency in the optimization process. Moreover, due to the relatively small proportion of positive samples compared to negative samples, the majority of anchors become redundant, resulting in a high computational cost. Conversely, anchor-free methods completely discard the concept of anchors, effectively avoiding the above problems and improving detection efficiency. For example, Densebox [20] can directly predict the center position and size information of objects, and then decode them to obtain the bounding box. Inspired by keypoints detection, some keypoint-based methods emerged. CornerNet [10] and ExtremeNet [21] first predict the key points at a specific position of objects and then encode them to obtain the final prediction result. Based on Cornernet, CenterNet [22] predicts the center position, object size, and offset to accurately determine the bounding boxes. Currently, anchor-free methods achieve competitive accuracy with considerable speed compared to anchor-based methods.

B. Deep Learning-Based Ship Detection Method in SAR Images

Since DCNN-based methods can automatically extract features without artificial preprocessing of SAR images, some methods specifically designed for SAR ship detection have been proposed and shown satisfactory performance.

Considering the variability of ship scale features, multiscale feature fusion methods have received a lot of attention. Fu et al. [23] design an LA module to emphasize structural information from different features and incorporate scale context through an adaptive saliency enhancement method. Zhang et al. [24] develop a BFPN module, which generates a set of unified-scale feature maps and employs a feature balance fusion method to capture object scale features. Jin et al. [25] utilize a series of dense blocks and an enlarged receptive field approach to enhance the contextual information of features. Furthermore, certain methods based on attention mechanisms and salient features are designed to increase the discriminability of ships in complex scenes. Sun et al. [3] design a category-position module to optimize the detector's regression task by generating guidance vectors from the features of the classification branch. These features for the regression task contain more positional information, enhancing the accuracy of target localization. Zhao et al. [26] introduce a visual enhancement module, which transforms the feature space to extract essential features in the frequency domain, to enhance the saliency of ships. Ke et al. [27] design an FEFPN that leverages the ship distribution of the dataset to enhance detection performance.

The above-mentioned method primarily employs HBB to locate ship targets. However, ships usually have large aspect ratios, which make it hard to visually convey their orientation and size information accurately using HBB. Moreover, the HBB representation typically contains a lot of redundant information, which makes it hard for the detector to extract ship features precisely. To address these issues, several SAR ship detection methods adopt OBB to represent detection results.

To address the discontinuity problem in orientation box prediction, Zhou et al. [12] encode the four endpoints of orientation boxes using ellipse parameters, which eliminate the need for orientation angle prediction. Li and Lang [13] employ inter-ship center point distances to filter out redundant and low-quality detection results, thereby improving the accuracy of orientation bounding boxes. Sun et al. [14] convert angle regression into angle classification, which effectively solves the regression inconsistency problem caused by sudden loss spikes. Zhang et al. [28] introduce a novel heatmap loss function that incorporates ship orientation angles and size information. This strategy reduces the position drift sensitivity, leading to a higher recall rate.

Despite the above methods achieving significant progress, they ignore the scattering characteristics and mainly focus on the contextual information around objects, making it challenging to achieve higher accuracy gains. To address these limitations, we construct a topological structure for scattering regions, allowing a broader receptive field to extract global information from the entire SAR image. Moreover, we employ contour information to enhance the saliency of ships and fuse scale information of objects to improve the robustness of multiscale ship detection.

III. Proposed Method

To enhance the performance of ship detection in complex scenes, an SRT-Net is constructed to dynamically select global

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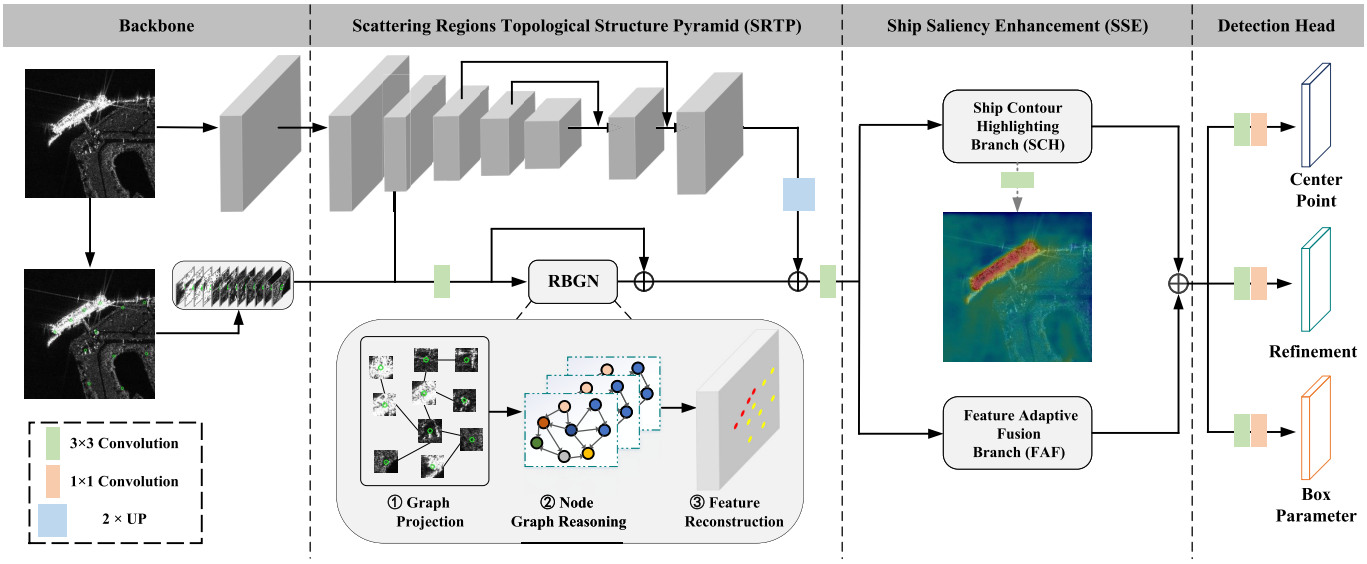


Fig. 3. Overview of the SRT-Net.

information from the entire image, leading to improved detection performance. The SRT-Net architecture is presented in Fig. 3, which is composed of four components: the backbone for feature extraction, the SRTP for global context extraction, the SSE module for SSE, and the detection head for final prediction. In this section, a detailed introduction to each component of the SRT-Net is provided. First, the primary anchor-free method and the presentation strategy for the OBB are introduced. Next, the SRTP is described in detail, which is used to adaptively extract the global context of scattering regions. Then, the SSE module consisting of two branches is described in detail, which guides the model to learn the contour information of objects. Finally, the SEA loss is introduced in the loss function part.

A. General Anchor-Free Baseline

Since ships are sparsely distributed in SAR images, anchor-based methods often generate numerous invalid anchor boxes, which can result in reduced accuracy and increased computation. To avoid the above problems, an anchor-free method named CenterNet [22] is used as our preliminary network structure. Its overall framework is shown in Fig. 4, and it has the following advantages in ship detection: it eliminates the anchor concept and uses keypoint detection to predict object locations, thereby reducing the number of required hyperparameters. Moreover, during the training process, it generates positive and negative samples based on ship center points, which helps to avoid the problem of densely arranged ships being easily overlooked due to excessive box overlap. Accordingly, our proposed method follows a similar structure to CenterNet, ensuring that densely packed ships are not overlooked while effectively reducing the amount of computation.

In the feature extraction stage, the proposed method is built on an encoding–decoding structure with ResNet50 [29] as the backbone for feature extraction and FPN [30] for feature fusion. ResNet50 utilizes five stacked downsampling convolutional blocks to capture features at different scales. The output feature maps of the last four blocks are marked as $\{C_2, C_3, C_4, C_5\}$, with corresponding downsampling rates and channel dimensions of $\{4, 8, 16, 32\}$ and $\{256, 512, 1024, 2048\}$, respectively.

The FPN is introduced to fuse these four features $\{C_2, C_3, C_4, C_5\}$, which integrates spatial information and semantic information through bilinear interpolation and skip connections. To ensure localization accuracy, the largest scale feature map Y' generated by FPN is used to predict the OBB. The feature Y' with rich semantic information is further through a 3×3 convolutional layer to generate the refined feature $Y \in \mathbb{R}^{(H/s)\times(W/s)\times 256}$. s denotes the downsampling stride, here $s = 4$, and $H \times W$ represents the size of the SAR image.

In the detection head, three branches are used to predict the center point map (CPM), refinement map, and box parameter map. The center point branch and refinement branch determine the precise position of ships. In addition, the box parameter branch is employed to predict the scale of ships. And each branch includes one 3×3 convolutional layer and one 1×1 convolutional layer.

The CPM $P \in \mathbb{R}^{(H/s)\times(W/s)\times 1}$ locates the center point of ships in SAR images. Assuming the center point of a ship object on the original SAR image is marked as (c_x, c_y), its corresponding position on P is denoted as (p_x, p_y), where $p_x = \lfloor c_x/s \rfloor$ and $p_y = \lfloor c_y/s \rfloor$. The center of each ship is considered a positive sample, whereas other positions on P are considered negative samples. Considering that adjacent points to positive samples may also generate considerable prediction results, a 2-D non-normalized Gaussian function $G(c_x, c_y, \sigma)$ [10] is employed to assign target values to the CPM P, as illustrated in Fig. 5

$$G\left(c_x, c_y, \sigma\right) = \exp\left(-\frac{\left(p_x - c_x\right)^2 + \left(p_y - c_y\right)^2}{2\sigma^2}\right) \tag{1}$$

where the parameter σ represents a scale-adaptive radius.

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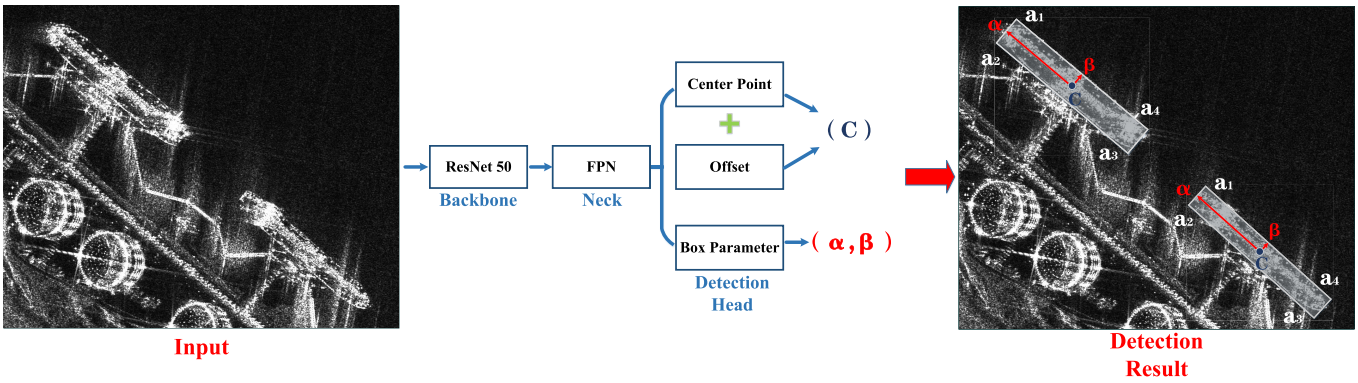


Fig. 4. Structure of the baseline method (CenterNet).

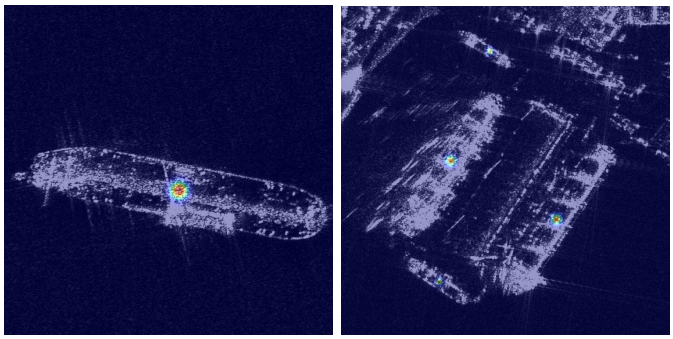


Fig. 5. Ground truth of CPMs.

The refinement map $R \in \mathbb{R}^{(H/s)\times(W/s)\times 2}$ assists in further refining the center of ships. In the feature extraction stage, the center point (c_x, c_y) on the original SAR image is mapped to the position $(\lfloor c_x/s \rfloor, \lfloor c_y/s \rfloor)$ on the P. However, in the decoding stage, it is difficult to accurately map the points on the P back to the original SAR image due to precision errors introduced by the downsampling stride. The refinement map is used to further refine the center position of the ships. The target value at position (c_x, c_y) is calculated as follows:

$$r_k = \left(\frac{c_x}{s} - \left\lfloor \frac{c_x}{s} \right\rfloor, \frac{c_y}{s} - \left\lfloor \frac{c_y}{s} \right\rfloor\right) \tag{2}$$

where r_k is the refinement value of the object k, and s is the downsampling rate.

The box parameter map $B \in \mathbb{R}^{(H/s)\times(W/s)\times 4}$ is used to predict the box parameters for the OBB. In Fig. 4, a vector definition scheme is introduced to encode OBBs. These boxes are constructed using the center point of the ship and the distances from the center point to the boundaries (a_1a_2 and a_1a_4). The distance is represented by a pair of mutually perpendicular vectors (α and β) that originate from the center point and extend to the edges of the bounding box. The element at the corresponding position on B represents the box parameters of the corresponding ship. Each element on the B is expressed as $(\alpha_x, \alpha_y, \beta_x, \beta_y)$, where (α_x, α_y) and (β_x, β_y) represent the coordinate representations of vectors α and β, respectively.

As depicted in Fig. 4, during the decoding stage, the four vertices of each bounding box can be represented as follows:

$$\begin{aligned} a_1 &= C + \alpha + \beta a_3 = C - \alpha - \beta \\ a_2 &= C + \alpha - \beta a_4 = C - \alpha + \beta \end{aligned} \tag{3}$$

where C represents the center position of a ship, determined by the CPM and refinement map.

B. Scattering Regions Topological Structure Pyramid (SRTP)

In SAR images, many artificial structures can exhibit similar visual characteristics and scattering mechanisms as ships, resulting in many false alarms. Additionally, imaging noise and strong scattering interference can decrease the prominence of ships, increasing the likelihood of missed detections. Therefore, more than relying on local descriptors for ship detection in complex scenes is required and the integration of global context is crucial. For instance, utilizing the object's position and contextual information from the surrounding background, the detector achieves more precise discrimination between ships and other targets, leading to a reduction in false alarms. Furthermore, leveraging the structural similarity among ships, the detector can enhance the saliency of ships and improve the detection performance.

To dynamically select the global context during feature fusion, a novel feature pyramid SRTP is designed to emphasize ships under strong scattering interference and reduce false alarms. This pyramid is based on the vanilla FPN [30] and features a specially designed region-based graph neural network (RBGN) module on the lateral connection. The RBGN module creates a topological structure of the entire original feature, providing a global context that supports object detection and recognition. Specifically, it utilizes GCNs to expand the receptive field of the model and extract more comprehensive semantic information from the entire image. Moreover, the RBGN module captures the feature similarities across each scattering region, allowing the incorporation of structural similarities into the features to enhance the prominence of ships.

The designed RBGN module, as depicted in Fig. 6, consists of three components: graph projection for establishing the topological structure of key regions, node graph reasoning for transferring global context, and feature reconstruction for generating enriched features with global context. Next, we will provide a detailed introduction to three components.

1) Graph Projection Stage: In single-polarization SAR images, SSRs contain more information for target detection

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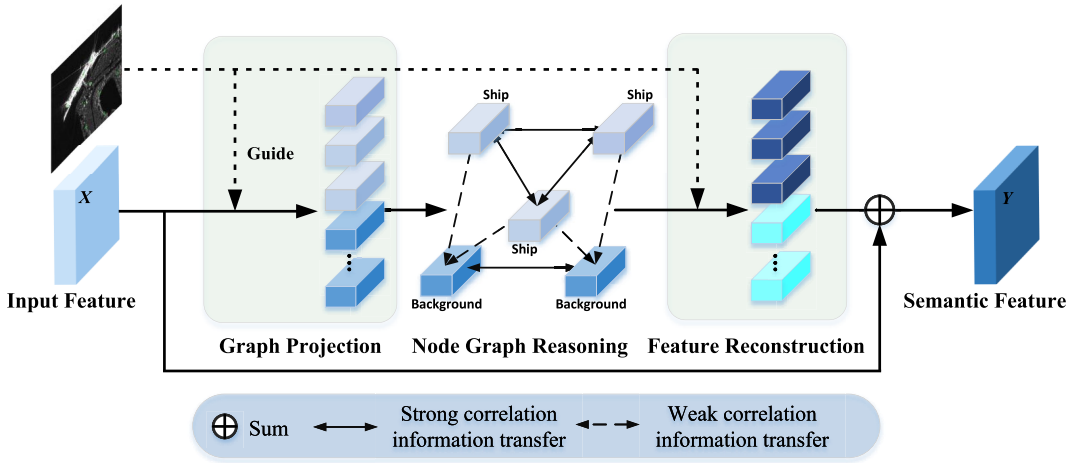
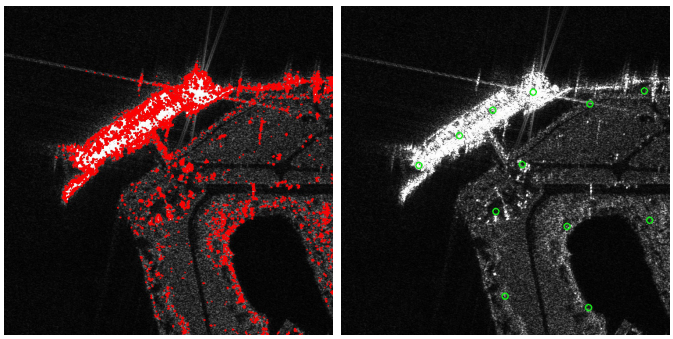


Fig. 6. Structure of the RBGN.



(a) (b)

Fig. 7. (a) Strong scattering points. (b) Scattering core points of the SSRs.

and recognition than weak scattering regions. We first identify and extract SSRs in the graph projection stage to take advantage of this information. These regions cover a large proportion of ships, artificial structures, and sources of strong scattering interference. They contain valuable information for establishing a topological structure based on the similarity of their features.

Specifically, we use the Harris corner point detector [31] to identify the raw strong scattering points in the SAR image, as illustrated in Fig. 7(a). This method can extract corner points that have local peak values and are invariant to scale and rotation. Therefore, it can effectively highlight the regions that are rich in key information, ensuring that the network does not excessively focus on background information with weak scattering intensities such as the sea surface. In order to capture the SSRs where the raw scattering points are clustered and avoid redundancy, we further cluster these points to reflect the distribution of scattering intensity. Based on the similarity between adjacent points, we use the k-means [32] method to group the irregular and redundant points into N clusters. N is a hyperparameter that determines the division of regions and is dependent on the spatial resolution of the SAR image. Each cluster represents an SSR, and these clusters cover most ships and artificial facilities in the SAR images. The centers of these clusters, shown in Fig. 7(b), are considered as the scattering core points of the SSRs. To preserve the information from the input feature, we select scattering core points as the fusion center and map similar features from the input feature to a set of nodes of the topological structure based on their similarity. Specifically, an assignment method is employed to convert the input feature into a graph structure that represents the topological structure. The graph structure is denoted as $G(V, R)$, where $V \in \mathbb{R}^{N\times 256}$ represents the nodes of the graph, and $R \in \mathbb{R}^{N\times N}$ represents the edges.

Here, we introduce the process of constructing the topological structure in detail. Specifically, we denote the input feature as $X \in \mathbb{R}^{(H/s)\times(W/s)\times 256}$, where $x_{ij} \in \mathbb{R}^{256}$ represents the feature vector at 2-D coordinates (i, j) of X. $P = \{P_1, P_2, P_3, \ldots, P_n, n = N\}$ represents a series of scattering core points (center points of clusters), each of which is represented by 2-D coordinates. Then, the scattering core point feature $F_p = (f_1, f_2, \ldots, f_N) \in \mathbb{R}^{256\times N}$ is extracted from corresponding position on X. Next, the variance of the column vector of F_p is calculated to derive the matrix $\sum = (\sigma_1, \sigma_2, \ldots, \sigma_N) \in \mathbb{R}^{256\times N}$, which is primarily employed for feature normalization.

Inspired by [33], we employ a soft assignment scheme to establish the topological structure. In this scheme, a_{ij}^k denotes the assigning weight of assigning feature vector x_{ij} to the feature vector of node k. The computation of the assigning weight can be expressed as follows:

$$a_{ij}^k = \frac{\exp\left(-\left\|\left(x_{ij} - f_k\right)/\sigma_k\right\|_2^2/2\right)}{\sum_i \exp\left(-\left\|\left(x_{ij} - f_i\right)/\sigma_i\right\|_2^2/2\right)} \tag{4}$$

where f_k represents the kth column vector of F_p, while σ_k corresponds to the kth column vector of Σ. The assigning weight a_{ij}^k is calculated based on the weighted Euclidean distance between x_{ij} and f_k, which serves as a measure of the feature similarity between the two vector features. The distance is then passed through a softmax function to create a soft assignment. The assigning weight matrix is denoted as $A \in \mathbb{R}^{H\times W\times N}$, with each element of A calculated using (4).

Once the assigning weight matrix A is computed, it can be used to map the input feature X to the node feature V of the topological structure. Let v_k represent the feature of the

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kth node in the topological structure. The value of v_k can be calculated as follows:

$$v'_k = \frac{1}{\sum_{ij} a^k_{ij}} \sum_{ij} a^k_{ij} \left(x_{ij} - f_k\right)/\sigma_k \tag{5}$$

$$v_k = \frac{v'_k}{\|v'_k\|_2} \tag{6}$$

where v'_k is a weighted average of the residuals between a feature vector x_{ij} and the scattering core point feature f_k. v_k is the feature vector of node k, obtained by further L2 normalization of v'_k. v_k represents the feature of the kth node, which also implies the feature of a connected SSR. Computing the feature vectors of all nodes in this way yields the node feature V of the topological structure.

The adjacency matrix R is constructed to form the edges of the topological structure and indicating the similarity of features across nodes. Specifically, the adjacency matrix R is generated by measuring the cosine similarity between each pair of node features. R is computed as follows:

$$R = \frac{V^T \times V}{\left\|V^T \times V\right\|} \tag{7}$$

where the symbol $\times$ represents matrix multiplication. The adjacency matrix R represents the similarity between each pair of nodes of the topological structure. And it is used to guide the message passing process in the subsequent stages of the network.

2) Node Graph Reasoning Stage: In the process of node graph reasoning, the GCNs are utilized to dynamically select the global context of the entire image and propagate global context across every node of the topological structure. Each node receives the necessary information from all other nodes and updates its features based on the adjacency matrix. By incorporating the global context of each SSR into the feature of each node, the model can effectively capture the inter-node relationships. Research [34] indicates that for most GCNs, a two- or three-layer model can achieve the best performance. Therefore, we design a cascaded two-layer graph convolutional model to achieve a balance between speed and performance. The input and output dimensions are kept consistent for each GCN. The output of the first layer of graph convolution is used as the input of the second layer of graph convolution to achieve node graph reasoning. Specifically, a single graph convolution operation can be defined with weight parameter $W \in \mathbb{R}^{256\times256}$ as follows:

$$\tilde{V} = f\left(RV^T W\right) \tag{8}$$

where $f(\cdot)$ represents the nonlinear activation function, such as ReLU, that is applied after the graph convolution operation. The weight parameter W is randomly initialized and then learned through the backpropagation algorithm for optimization. $\tilde{V} \in \mathbb{R}^{256\times N}$ represents the new node feature that is generated by the graph convolution network. The enhanced node feature fuses the similarity information from each SSR, providing richer semantic information.

3) Feature Reconstruction Stage: During the process of constructing the topological structure, an assigning weight matrix A is utilized to map the input feature X to the node feature V. To map the enhanced node feature back to the original coordinate space, A is reused to re-weight the node features. The feature reconstruction operation, which generates global context-enhanced features, is as follows:

$$\hat{F} = A\tilde{V}^T + F, A \in \mathbb{R}^{H\times W\times N} \tag{9}$$

where $\hat{F} \in \mathbb{R}^{(H/s)\times(W/s)\times256}$ is the enhanced feature fused with the global context.

Compared to vanilla FPN, the SRTP constructs a topological structure, allowing a broader receptive field to extract global context from the entire images in an adaptive manner, enhancing ships' salience and suppressing false alarms. The topological structure of SRTP enables the learning of weak correlations between ship and artificial structure nodes through node-to-node information transfer, improving the separability of ships from backgrounds and effectively reducing false alarms. Moreover, when both nodes contain ship features, the RBGN module utilizes their strong correlation to highlight the semantic characteristics of ships leading to improved localization accuracy.

C. SSE Module

In SAR images, the existence of imaging noise and strong scattering interference brings challenges to the precise localization of ships. First, the strong scattering interference of artificial facilities blurs the contour information of the ship, preventing the detector from judging the edge of ships. Additionally, the speckle noise of SAR images may obscure low-scattering parts of ships, such as the deck. These interferences significantly increase the difficulty of predicting the size and contour information of ships. However, the head module of the current mainstream detection method mainly focuses on the positions of the four vertices, ignoring the pixel-level learning of positive and negative samples of the image. Therefore, these methods cannot accurately determine whether the high scattering intensity pixels around the ship belong to the ship or the background, leading to numerous inaccurate localizations. To solve the above issues, we propose an SSE module, which guides the model in learning the geometric feature information of ships to improve the detector's ability to capture ship contours accurately. Fig. 8 illustrates the structure of the SSE module, which is comprised of two branches: the ship contour highlighting branch (SCH) and the feature adaptive fusion branch (FAF).

1) SCH Branch: The SCH branch aims to highlight the geometric information of ships to enhance their saliency under various imaging conditions. A novel auxiliary branch is added to the network to guide the model to extract ship contour information. And the contour highlighting map (CHM) is devised to enable the network to accurately identify ship contours at a pixel level, thereby enhancing the localization accuracy in complex scenes. As depicted in Fig. 8, $X \in \mathbb{R}^{(H/s)\times(W/s)\times256}$ is the output of the SRTP, which fuses multiscale features. X is first fed into three 3×3 convolution to obtain a more refined

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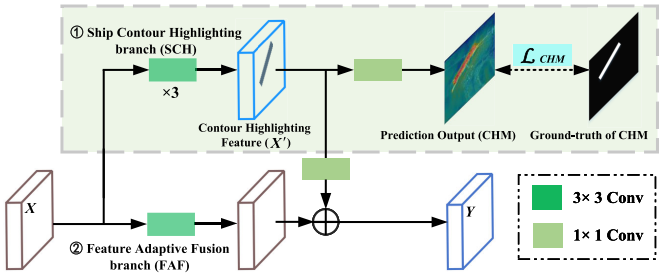


Fig. 8. Structure of the SSE module.

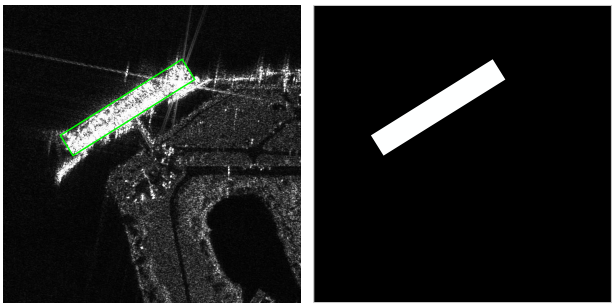


Fig. 9. (a) Ground-truth bounding box. (b) Ground truth of CHM.

feature map $X' \in \mathbb{R}^{(H/s)\times(W/s)\times 256}$. Then, the X' is utilized to predict CHM $C_{\text{CHM}} \in \mathbb{R}^{(H/s)\times(W/s)\times 1}$ through one 1×1 convolution. The CHM represents the contour information of ships, which is automatically generated by the expert annotation. For each point within the ground-truth bounding box of the input image, the target value of the corresponding position on CHM is set to 1, while the other positions are set to 0. As shown in Fig. 9, each pixel of CHM is labeled as a ship or background, providing guidance for the network to engage in pixel-level learning and accurately capture ship contour details. The SCH branch uses the backpropagation algorithm to guide the learnable parameters of the model to update in the set direction. This supervised classification task promotes the parameters of three 3×3 convolution with a more powerful ability to highlight the detailed information. Therefore, the ship contour information of X' is highlighted. Additionally, the model can gain a better understanding of the overall ship structure through the learning model of ship geometric information. It effectively prevents the low scattering intensity part of the ship from being overlooked due to strong scattering interference, and mitigates the risk of mistakenly identifying a single ship target as multiple independent targets.

2) FAF Branch: The FAF branch is designed to dynamically fuse the original feature and contour highlighting feature, allowing for the full utilization of spatial and semantic information from both features in an adaptive manner. As shown in Fig. 8, we use one 1×1 convolution to achieve a cross-channel feature fusion of the contour highlighting feature. The output of the FAF branch is computed as follows:

$$X_{\text{CTR}} = \text{con}v_{3\times 3}(X) + \text{con}v_{1\times 1}(X') \tag{10}$$

where X_{CTR} is the output of the FAF branch, $\text{con}v_{3\times 3}$ is a 3×3 convolution, and $\text{con}v_{1\times 1}$ is a 1×1 convolution.

The SSE module introduces a ship contour supervision task and a simple feature fusion strategy, guiding the detector to learn ship contour information and structural features, enhancing the saliency of ships in various imaging conditions.

D. Loss Function

The loss function of the training process can be divided into four parts:

$$\mathcal{L} = \mathcal{L}_{\text{cp}} + \mathcal{L}_{rf} + \mathcal{L}_{\text{box}} + \lambda \mathcal{L}_{\text{CHM}} \tag{11}$$

where $\mathcal{L}_{cp}$, $\mathcal{L}_{rf}$, $\mathcal{L}_{\text{box}}$, and $\mathcal{L}_{\text{CHM}}$ represent the center point loss, the refinement loss, the box parameter loss, and the ship contour loss, respectively. λ is a loss weight parameter for ship contour loss, and it is set to 0.5 in all our experiments.

In the part of the prediction CPM, to alleviate the problem of positive and negative sample imbalance, $\mathcal{L}_{cp}$ is derived from focal loss [19], which can be defined as follows:

$$\mathcal{L}_{cp} = -\frac{1}{N}\sum_{x,y}\begin{cases}(1-p_{xy})^{\alpha}\cdot\log(p_{xy}), & \text{if } \hat{p}_{xy}=1\\ (1-\hat{p}_{xy})^{\beta}\cdot p_{xy}^{\alpha}\cdot\log(1-p_{xy}), & \text{otherwise}\end{cases} \tag{12}$$

where p_{xy} is the predicted ship center point position, and $\hat{p}_{xy}$ is the actual ship position on the CPM. N refers to the number of ship samples in total. α is a weight parameter representing the degree of attention to difficult samples. As a balancing parameter, β assists in balancing positives and negatives ($\alpha = 2$ and $\beta = 4$ in our method).

The $\mathcal{L}_{rf}$ is the refinement loss, which is optimized using smooth L1 loss

$$\mathcal{L}_{rf} = \frac{1}{N}\sum_{k=1}^{N}\text{smooth}_{L1}(r_i - \hat{r}_i) \tag{13}$$

where $\hat{r}_i$ represents the localization error of ship i and r_i is the predicted refinement value of the same location.

The $\mathcal{L}_{\text{box}}$ is the box parameter loss. And we also utilize smooth L1 loss to optimize the box parameter

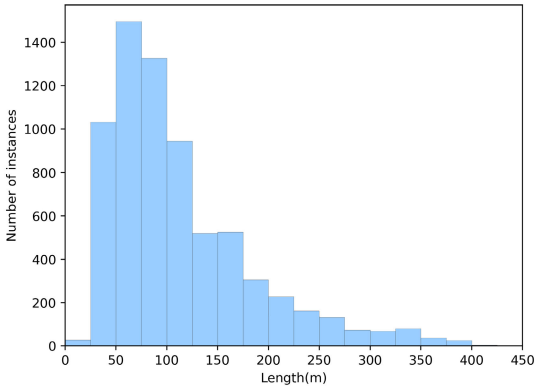
$$\mathcal{L}_{\text{box}} = \frac{1}{N}\sum_{k=1}^{N}\text{smooth}_{L1}(B_i - \hat{B}_i) \tag{14}$$

where $\hat{B}_i$ represents the actual scale information of the ship i and B_i is the predicted box parameter of the same object.

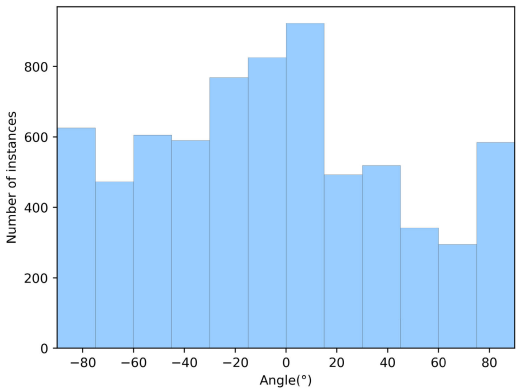
The $\mathcal{L}_{\text{CHM}}$ is the ship contour loss. When the vanilla focal loss is utilized to supervise the CHM $C_{\text{CHM}} \in \mathbb{R}^{(H/s)\times(W/s)\times 1}$, large ships contribute significantly more to training loss than small ships. Therefore, small ships are easily missed due to the imbalance of scale information. Inspired by [35], the SFA loss is introduced to provide a means of balancing the multiscale characteristics of ships and alleviating the problem of unbalanced attention caused by the differences in scale. This method utilizes the size of each ship to normalize the corresponding loss and can be described as follows:

$$\mathcal{L}_{\text{CHM}} = -\frac{1}{N}\sum_{x,y}\begin{cases}\frac{1}{S_{xy}}(1-c_{xy})^{\alpha}\cdot\log(c_{xy}), & \text{if } \hat{c}_{xy}=1\\ c_{xy}^{\alpha}\cdot\log(1-c_{xy}), & \text{otherwise}\end{cases} \tag{15}$$

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(a)



(b)

Fig. 10. (a) Distribution of ship length in LSSDD. (b) Distribution of ship orientation in LSSDD.

where $\hat{c}_{xy}$ denotes the target value at position (x, y) on the CHM, while the predicted value at the same location is represented by c_{xy}. S_{xy} represents the size of the ship sample at the positive sample point (x, y), and it can be determined by the total number of grid points inside the OBB. N represents the grid point number of positive samples in CHM. By normalizing the loss with the ship size, the SFA loss can improve the focus on small-scale ships and enhance the detector's robustness to multiscale ship detection.

IV. Experiments

A. Dataset and Settings

To validate the effectiveness of our proposed method, we conduct a series of experiments on a large-scale ship dataset called LSSDD. Additionally, to verify the generalization and robustness of the model, we introduce the public SSDD+ dataset for further comparative experiments. Below is the relevant information about the dataset:

1) LSSDD: The LSSDD contains 116 single-polarization SAR images from the Gaofen-3 satellite annotated by experts. These images cover 16 complex port scenes and provide 6982 multiscale and multioriented labeled SAR ship instances. Fig. 10 shows the distribution of ship length and orientation. Furthermore, LSSDD offers a spatial resolution of 1 m and sizes ranging from 12 000 × 12 000 to 40 000 × 40 000 pixels. A total of 94 SAR images are randomly selected for the training and validation set, and 22 others are chosen for the test set. The same training set is employed for all experiments, as well as the same test set for comparing the effects.

2) SSDD+: The SSDD+ dataset [36] is a refined annotation of the SSDD dataset with OBB. It contains multiscale ship slices from RADARSAT-2, TerraSAR-X, and Sentinel-1 satellites, with a total of 1160 images and 2456 annotated ships. The image resolution varies from 1 to 15 m. The SSDD+ dataset is divided into training, validation, and test sets in a ratio of 7:1:2.

B. Implementation Details

All experiments of our proposed method use ResNet50 as the backbone. Each batch of eight images is run through 100 epochs on a 16 GB NVIDIA Tesla P100 GPU. To increase generalization abilities and avoid overfitting, we randomly cropped the raw large-scale SAR images of the training dataset into patches of different scales (800×800, 1024×1024, and 1300×1300) while ensuring that each patch contained the ship object. A total of 10 214 patches are obtained in this manner, and then resized to 608×608 for training. To converge the loss, we apply the Adam optimizer, starting with an initial learning rate of 0.000125. After 70 epochs, the learning rate is then reduced to 0.0000125.

Additionally, the original SAR images for the test set are cropped into 1024×1024 pixel patches with 512 pixel overlaps by the mode of sliding windows. During the inference stage, each patch is fed to the detector to obtain individual detection results and then these detection results are merged to generate the final detection result for large-scale images. To remove overlapping bounding boxes, the NMS [37] is applied during the merge process with IoU thresholds of 0.2.

C. Evaluation Metrics

Four commonly used evaluation criteria are employed to assess the accuracy and completeness of the ship detection task: precision, recall, F1 score, and average precision (AP). First, to evaluate the quality of the predicted bounding box, we calculate the IoU of each predicted result $\mathrm{B_p}$, which represents the overlap area between $\mathrm{B_p}$ and the matched ground-truth bounding box $\mathrm{B_{gt}}$

$$\mathrm{IoU} = \frac{\mathrm{area}\left(\mathrm{B_p} \cap \mathrm{B_{gt}}\right)}{\mathrm{area}\left(\mathrm{B_p} \cup \mathrm{B_{gt}}\right)}. \tag{16}$$

Based on the IoU the detection results can be divided into two categories: true positives (TP) and false positive (FP). As a general rule, prediction results with an IoU greater than 0.5 are regarded as TP, while those with an IoU less than 0.5 are considered FP. In addition, FN refers to ships that fail to match detection results. Precision and recall are used to compute the accuracy and completeness of detection results, respectively. They are computed as

$$\mathrm{Precision} = \frac{\mathrm{N_{TP}}}{\mathrm{N_{TP}} + \mathrm{N_{FP}}} \tag{17}$$

$$\mathrm{Recall} = \frac{\mathrm{N_{TP}}}{\mathrm{N_{TP}} + \mathrm{N_{FN}}} \tag{18}$$

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TABLE I
IMPACT OF EACH COMPONENT OF SRT-NET

SRTP	SSE Module	Recall	Precision	F1	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.8}$
×	×	0.7163	0.7376	0.7268	0.7410	0.3397	0.1893
✓	×	0.7772	0.7951	0.7861	0.7877	0.4131	0.2307
×	✓	0.7788	0.7618	0.7702	0.7924	0.4144	0.2511
✓	✓	**0.8141**	**0.8038**	**0.8089**	**0.8271**	**0.4370**	**0.2786**

where N_{TP}, N_{FP}, and N_{FN} represent the total number of TP, FP, and FN respectively. Furthermore, the F1 score is introduced for comprehensively evaluating a model's accuracy and completeness

$$\mathrm{F1} = \frac{2 \times \mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{19}$$

To evaluate the overall performance of detectors, the AP metric is commonly used. The detection results are sorted in descending order based on the confidence scores. By varying the confidence score threshold, different pairs of precision and recall can be obtained. AP stands for average precision under varying recall rates $Q = \{0, 0.01, \ldots, 1\}$

$$\mathrm{AP} = \frac{1}{101} \sum_{q \in \mathrm{Q}} \mathrm{Precision}|\mathrm{Recall} = \mathrm{q}. \tag{20}$$

Equation (20) shows the formula for calculating the AP, where the IoU threshold is set to 0.5, and the metric is denoted as $AP_{0.5}$. To better evaluate the localization accuracy of the detector, we use more stringent IoU thresholds of 0.75 and 0.8, and the metrics are represented by $AP_{0.75}$ and $AP_{0.8}$, respectively.

D. Ablation Studies

A series of ablation experiments are carried out on the LSSDD dataset to examine the contributions of each design component and the overall performance of the proposed method. To ensure a fair comparison, we maintain consistent settings for all experiments. The original detection network is used as the baseline. The two proposed modules are gradually added to the original network, and Table I presents the overall results. Each of our proposed methods significantly improves detection performance. Compared to the baseline, our final network achieves significant improvements in AP at different IoU thresholds. Specifically, it improves by 8.61%, 9.73%, and 8.93% on $AP_{0.5}$, $AP_{0.75}$, and $AP_{0.8}$, respectively. In addition, our recall and precision have improved by 9.78% and 6.62%, respectively. These results clearly demonstrate that the proposed components enhance the localization accuracy of ship objects and reduce false alarms. Next, we will analyze the contribution of each module in detail.

1) Effect of SRTP: A topology network of the proposed detection neck guides the detector to focus on SSRs and dynamically capture a comprehensive global context across key regions. We first select different numbers of SSRs (4,8,12,16) for comparison experiments. In Table II, it can be observed that the introduction of SSR gains steady improvement compared with the baseline, and the model achieves the best accuracy when the number of SSR is 12. Hence, it is essential to select an appropriate number of SSR. When an insufficient number of SSR are selected, it may fail to adequately cover key areas, making it challenging to exclude false alarms and potentially overlooking small ship objects. Conversely, an excessive number of SSR may result in ship objects being divided into multiple parts, leading to incomplete detection or omission. And it will also bring unnecessary computational burden. In our experiments, we design the number of SSR as 12, achieving a good balance in effectively covering the SSRs and the computational complexity.

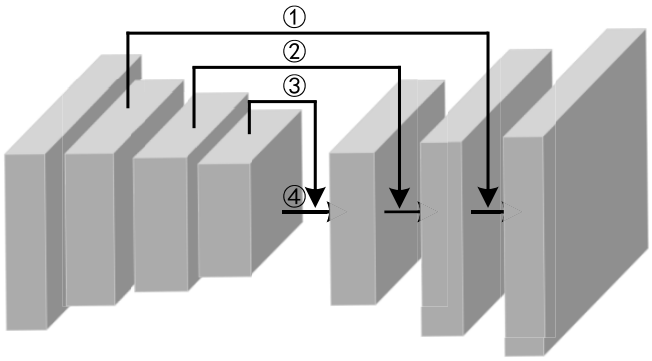


Fig. 11. Comparison of the influence of different positions of the RBGN module.

As shown in Fig. 11, we analyze the performance when the designed RBGN module is added to different lateral connection layers of the vanilla FPN. Results in Table III show that adding RBGN to different lateral connection layers impacts the detection performance and the best performance is achieved when RBGN is added to the first lateral connection layer of the FPN. This is because, for larger feature maps, the guided localization of SSRs achieves higher accuracy.

Table I shows that the SRTP achieved a considerable improvement of 4.67% in $AP_{0.5}$ compared to the baseline. Moreover, SRTP significantly enhances the precision metric, indicating its effectiveness in reducing false alarms. The comparison results are visualized in Fig. 12. In Fig. 12(a), the baseline method misidentifies buildings with similar texture features as ships, and it fails to capture some offshore ships under strong scattering interference. In contrast, Fig. 12(b) demonstrates that the method with the SRTP effectively highlights ship objects while efficiently reducing false alarms. The detection result clearly shows that the global context provided by SRTP enhances the salience of the ship, effectively

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TABLE II
IMPACT OF DIFFERENT NUMBERS OF SSR

the number of SSR	Recall	Precision	F1	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.8}$
4	0.7420	0.7384	0.7402	0.7539	0.3753	0.2150
8	0.7596	0.7783	0.7714	0.7744	0.3837	0.2225
12	0.7772	**0.7951**	**0.7861**	**0.7877**	**0.4131**	**0.2307**
16	**0.7837**	0.7677	0.7756	0.7812	0.3998	0.2232

TABLE III
IMPACT OF DIFFERENT POSITIONS OF SRTP

Position	Recall	Precision	F1	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.8}$
1	**0.7772**	**0.7951**	**0.7861**	**0.7877**	**0.4131**	**0.2307**
2	0.7580	0.7805	0.7691	0.7806	0.4049	0.2288
3	0.7436	0.7557	0.7496	0.7682	0.3631	0.2005
4	0.7388	0.7508	0.7447	0.7564	0.3559	0.2136

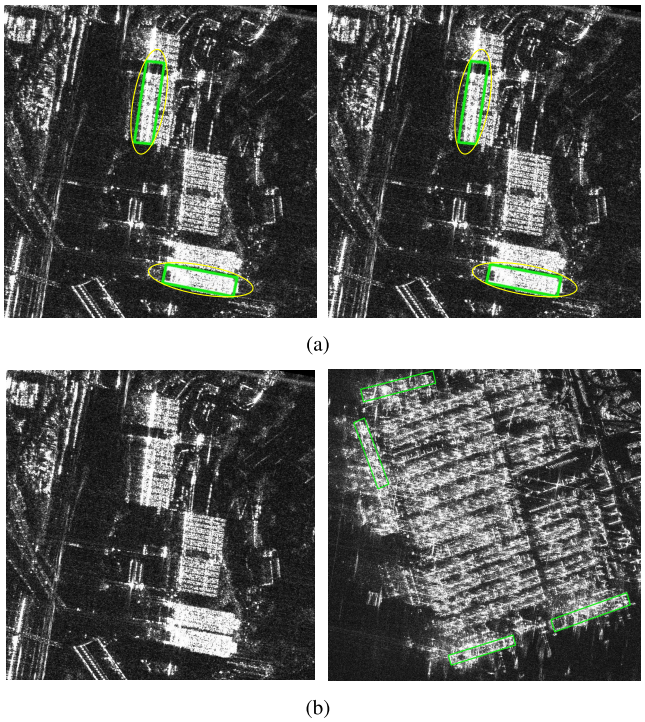


Fig. 12. Visualization comparison methods without and with the SRTP. Detection results are marked in green boxes, false alarms are marked in yellow circles, and missing ships are marked in red circles, respectively. (a) Detection results of the baseline without SRTP. (b) Detection results of the baseline with SRTP.

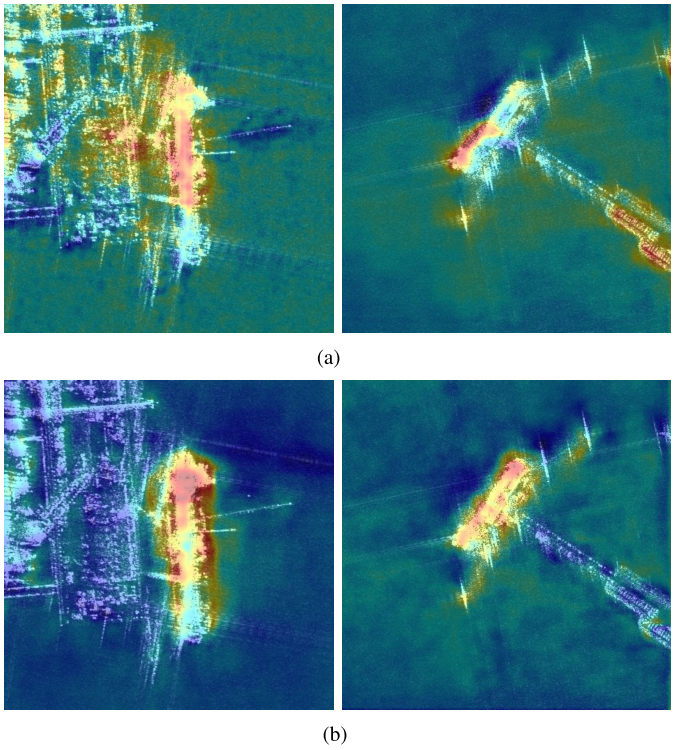


Fig. 13. Visualization of the features extracted by different methods. (a) Output features of the original FPN. (b) Output features of SRTP equipped with RBGN.

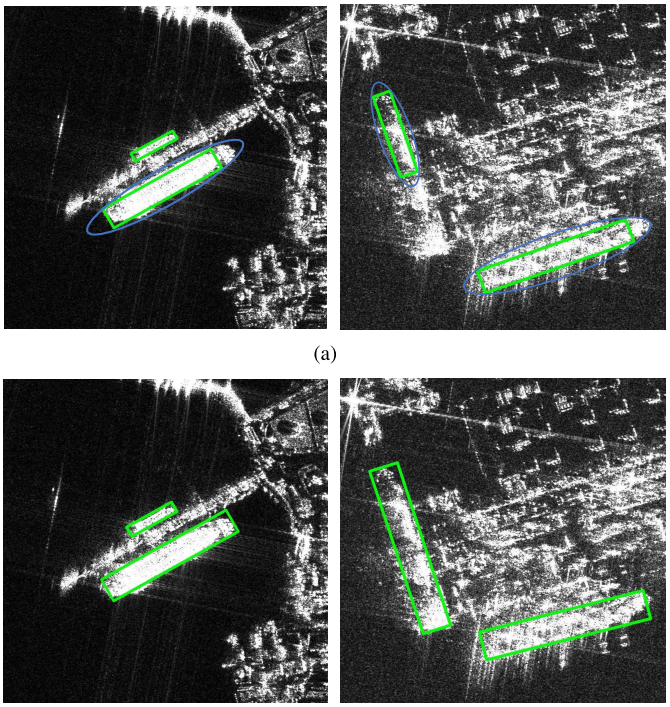
suppressing complex background interference and exhibiting a robust anti-interference capability.

To better demonstrate the effectiveness of RBGN, we visualize the output feature maps of the original FPN (baseline method) and SRTP equipped with RBGN. For the output feature maps, we compute the sum of elements at each position to generate an activation heatmap for visualization. As shown in Fig. 13(a), due to the similar visual properties and scattering characteristics between some inland regions and ships, several artificial facilities have higher activation values. This makes it difficult for the detector's attention to focus on the ship target in complex scenes, which can cause inaccurate localization and false alarms. To address these issues, we add an RBGN module in the feature fusion stage to establish a topological structure for scattering regions and transmit global context to support target detection and recognition. In Fig. 13(b), the ship target becomes more salient, and the global context captured by the RBGN module significantly suppresses the activation values of artificial facilities. It proves that the RBGN

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TABLE IV
ABLATION STUDIES OF SSE MODULE

SCH branch		FAF branch	Recall	Precision	F1	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.8}$
supervised	unsupervised							
×	×	×	0.7163	0.7376	0.7268	0.7410	0.3397	0.1893
×	✓	✓	0.7452	0.7346	0.7399	0.7501	0.3591	0.2082
✓	×	×	0.7724	0.7531	0.7627	0.7834	0.3954	0.2331
✓	×	✓	**0.7788**	**0.7618**	**0.7702**	**0.7924**	**0.4144**	**0.2511**



(a)

(b)

Fig. 14. Visualization comparison methods without and with the SSE module. Detection results are marked in green boxes; objects with low localization accuracy are marked in blue circles. (a) Detection results of the baseline without SSE module. (b) Detection results of the baseline with SSE module.

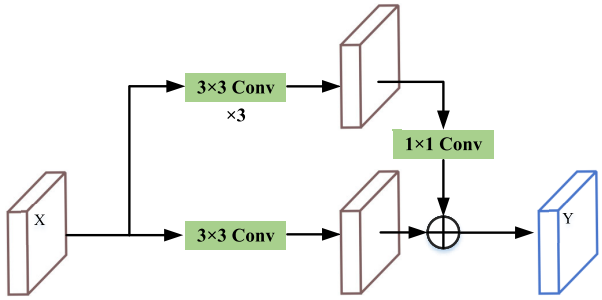


Fig. 15. SSE module structure without ship contour supervision.

module, by providing a larger receptive field, enables the reconstructed features to have richer semantic information, which can enhance the saliency of ships. Moreover, we employ GCNs to propagate feature similarity across SSRs, which can improve the separability of ships and artificial facilities in an adaptive manner.

2) Effect of SSE Module: Table I shows that the SSE module improves the $AP_{0.5}$, $AP_{0.75}$, and $AP_{0.8}$ by 5.14%, 7.47%, and 6.18%, respectively, compared to the baseline. The effectiveness of the SSE module is further demonstrated in Fig. 14. In Fig. 14(a), the baseline method struggles to capture the overall structure of the ship in complex scenes, leading to inaccurate localization. Especially for ships docked at ports, their contour information can be easily obscured by the strong scattering interference of surrounding artificial facilities, which hampers the detector's ability to distinguish ship edges. Conversely, Fig. 14(b) demonstrates that the method equipped with the SSE module can more precisely locate ships. This indicates that the SSE module enhances the saliency of ship targets in complex scenes, and effectively guides the detector to learn the ship contours and structural features, thereby improving the accurate localization of the targets.

The SSE module consists of two branches: one dedicated to highlighting ship contour information in complex scenes, and the other for adaptive feature fusion. To investigate the impact of the ship contour supervision component of the SCH branch, an experiment was conducted where the ship contour supervision was removed while keeping the same network structure, as depicted in Fig. 15. Table IV demonstrates that the ship contour supervision component significantly improves the $AP_{0.5}$ metric and recall rate, indicating that the SCH branch effectively enhances the saliency of ships in complex scenes. Moreover, the SCH branch significantly enhances the $AP_{0.75}$ metric, indicating that it enables accurate extraction of ship contour information for precise localization. The FAF branch also plays a positive role, achieving further gains in detection performance.

Furthermore, the impact of the SFA loss is also verified. We performed the corresponding ablation study for SFA loss in Table V. Compared with the vanilla focal loss, the SSE module can achieve 1.59% higher $AP_{0.5}$ when using the SFA loss. Furthermore, Fig. 16 displays the detection results obtained with two different supervised strategies. Fig. 16(a) reveals that when the vanilla focal loss is employed, some small ships are easily overlooked. Conversely, Fig. 16(b), which utilizes the SFA loss, demonstrates effective detection of multiscale ship detection. It is evident from these results that the SFA loss enhances attention toward small-scale ships and exhibits excellent detection capability.

The ship contour loss $\mathcal{L}_{\text{CHM}}$ in (11) is weighted by a loss weight parameter λ. Table VI shows that parameter λ has a

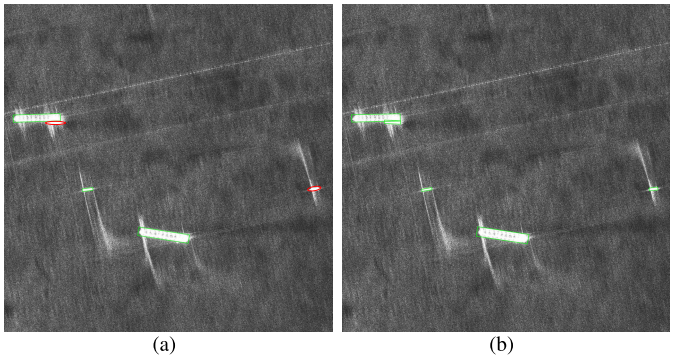
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TABLE V
ABLATION STUDIES OF SEA LOSS

Loss	Recall	Precision	F1	$AP_{0.5}$
Focal Loss	0.7548	0.7512	0.7530	0.7765
SFA Loss	**0.7788**	**0.7618**	**0.7702**	**0.7924**

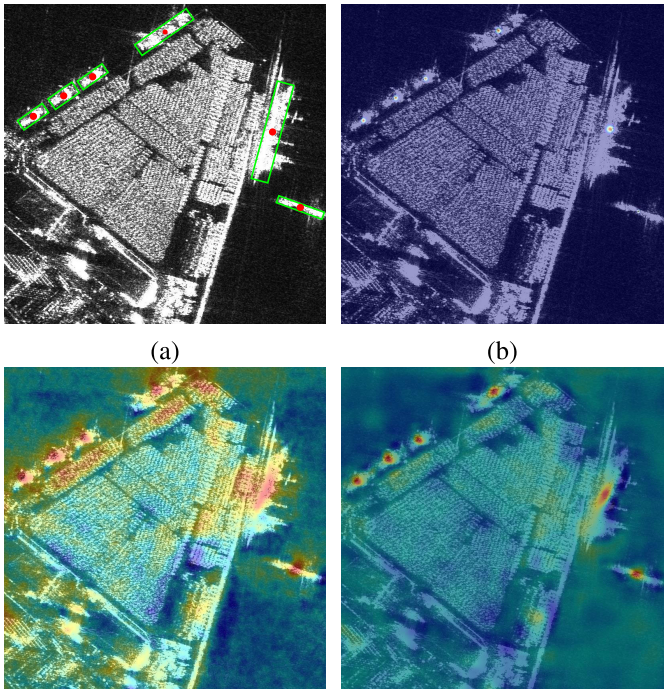
TABLE VI
ABLATION STUDIES OF PARAMETER λ

λ	0.3	**0.5**	0.8	1	2
$AP_{0.5}$	0.7883	**0.7924**	0.7876	0.7895	0.7863



(a) (b)

Fig. 16. Visualization comparison of different loss functions for the SSE module. Detection results are marked in green boxes; missing ships are marked in red circles. (a) Detection results with focal loss. (b) Detection results with SFA loss.



(a) (b) (c) (d)

Fig. 17. Visualization of the CPMs. Ground truths are marked in green boxes; the red dots indicate the center point positions of the ships. (a) Ground truth with labeled center point positions. (b) Ground truth of CPMs. (c) Visualization of the baseline without SRTP. (d) Visualization of the baseline with SRTP.

limited effect on the detection performance, only marginally affecting $AP_{0.5}$ by 0.5%. When the parameter λ is set to 0.5, all the losses reach a good balance.

E. Discussion

1) Analysis of the SRTP: Table I presents the contribution of the SRTP. To visually demonstrate the effect of the model and its practical application ability in complex scenes, we visualize the CPM from the detection head. The value of each pixel in CPM directly reflects the confidence score that the detector assigns to this point being the center point of the ship, ranging from 0 to 1. Specifically, the CPM is resized to match the input image. We multiply each value by 255 and convert it into a color image. The positions with lower confidence scores are closer to blue (the detector considers them more likely to be background), while the positions with higher confidence scores are closer to red (the detector considers them more likely to be the center of ships).

In Fig. 17(a), a series of containers on the dock exhibit high confidence scores, presenting large regions of red and yellow, while they should be shown in blue. It shows that in the absence of global context, the detector faces difficulty in making accurate judgments on objects with similar texture features, and it is easy to misidentify background targets as ships. Moreover, the baseline method shows excessive red areas at the ship center point location, which even spread to the ocean surface. It indicates that the baseline method is difficult to accurately extract the center point in complex scenes, and precise localization is still challenging. On the other hand, Fig. 17(d) shows that inland areas have changed from yellow and red to blue, which means that scores for artificial facilities are significantly reduced by SRTP's effect. Moreover, the designed SRTP can guide the detector to precisely identify the center point of ship. The predicted center points are closer to the center of ships, enabling high-precision localization. Two factors can be attributed to the improvement brought by SRTP. First, SRTP conveys similarity information across SSRs, enhancing the recognition ability of these regions and reducing false alarms. Second, the introduction of global context from SRTP effectively highlights the semantic features of ships under strong scattering interference to enhance localization accuracy.

1) Analysis of SSE Module: The proposed SSE module aims to provide additional guidance for detector to accurately locate ship contours under complex imaging conditions. The improvement achieved by each branch is illustrated in Table IV, and Fig. 18 shows the visualization results of CPM with and without the SSE module. In Fig. 18(c), some parts of the harbor (such as the lower right and upper areas of the ship) have large yellow and red regions, indicating that the detector tends to misclassify them as ships. Moreover, because of the close docking of ships at the port, the baseline method has a large red region at the center point of the ship, indicating that the baseline method has difficulty in locating the center point accurately. On the other hand, in Fig. 18, confidence scores related to artificial facilities are significantly reduced and shift to blue color. At the same time, center point positioning of ships is more accurate. It is clear that using SSE module

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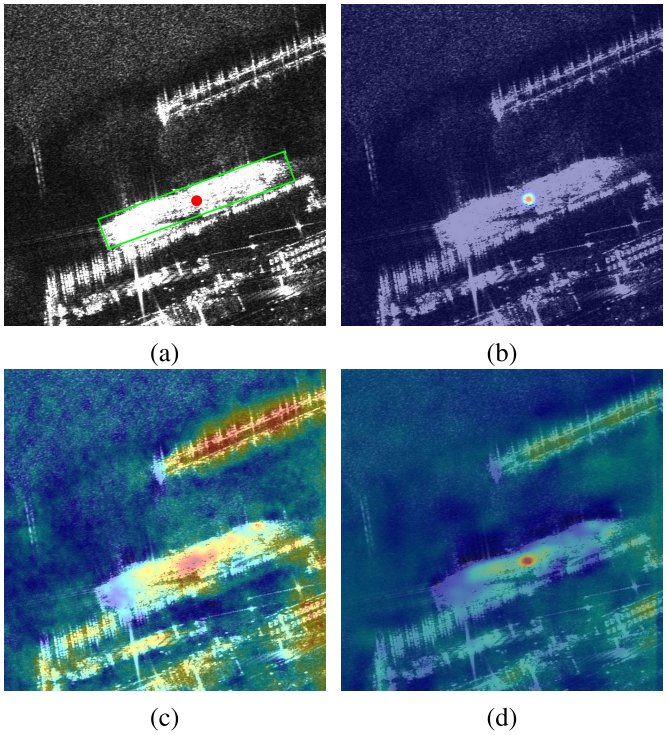


Fig. 18. Visualization of the CPMs. Ground truths are marked in green boxes; the red dots indicate the center point positions of the ships. (a) Ground truth with labeled center point positions. (b) Ground truth of CPMs. (c) Visualization of the baseline without SSE module. (d) Visualization of the baseline with SSE module.

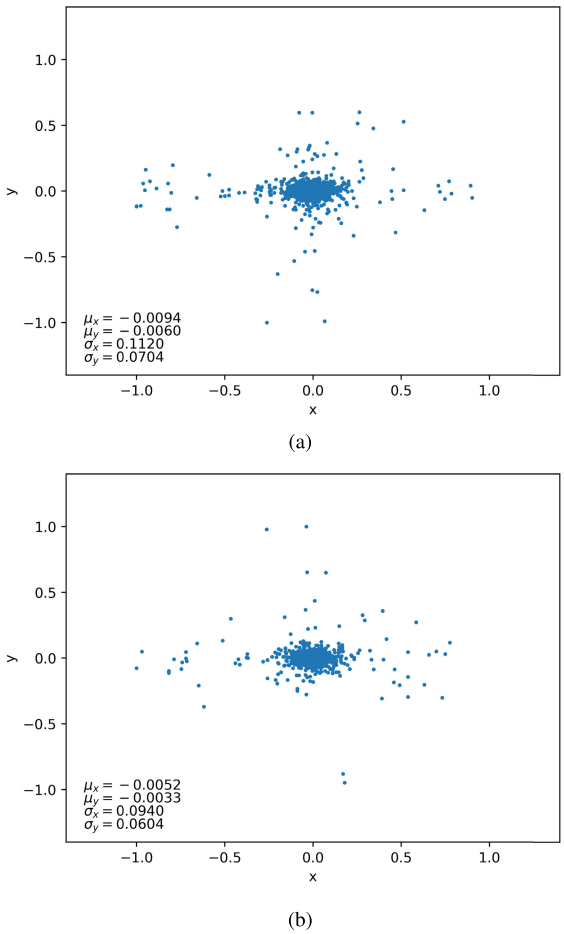


Fig. 19. Localization error comparison methods without and with the SSE. (a) Localization error of the baseline without SSE module. (b) Localization error of the baseline with SSE module.

can accurately locate center points and effectively suppress interference.

To better illustrate the role of the SSE module in improving positioning accuracy in complex scenes, we evaluate the localization error. The experimental results are presented in Fig. 19. Specifically, we first match the predicted box with the ground-truth box based on IoU. Then, we calculate the distance between the four endpoints of the best-matched predicted box and ground-truth box. Considering that ship size has a significant impact on localization errors, we normalize the localization errors according to the size of the target. The average values of the localization error calculated along the horizontal and vertical axes are (μ_x, μ_y), with standard deviations of (σ_x, σ_x). From Fig. 19, it is evident that SSE module can significantly improve target localization accuracy compared to baseline. This indicates that SSE module effectively enhances the saliency of ships in complex scenes, enabling detectors to better capture ship contour information and geometric features, thereby improving subsequent detection tasks' localization accuracy.

There are three potential reasons for the observed improvements. First, the SCH branch of SSE module utilizes the backpropagation algorithm to highlight the ship contour information within the features, helping the detector better capture the ship target and accurately locate its position. Second, the pixel-level classification task for the ship's overall structure enhances the detector's sensitivity to the ship's geometric information, which effectively prevents the misidentification of other objects with similar shapes as ships. Third, the feature fusion method enables the adaptive dynamic selection of contextual features, which fully utilizes the spatial and semantic information of the two features at different stages.

F. Comparison With CNN-Based Methods

To better show the performance of SRT-Net, we compare the proposed detector with other CNN-based methods. We select nine methods, including R^2CNN [38], Oriented R-CNN [39], Rotated ATSS [40], RoI Transformer [41], S^2A-Net [42], Oriented RepPoints [43], Oriented FCOS [44], Oriented CenterNet [22], SKG-Net [45], for a series of comparative experiments on LSSDD, covering both anchor-based and anchor-free methods. The SKG-Net introduces a feature adaptation method which uses the SKG strategy to provide additional guidance for convolution to obtain stronger feature extraction ability for ship detection. The comparative experiments on LSSDD show the detection results in Table VII. Our method achieves 82.71% $AP_{0.5}$ and 80.89% F1 on LSSDD, and also outperforms all other methods in both precision and recall. Additionally, the $AP_{0.75}$ and $AP_{0.8}$ are also improved to some degree. It indicates that our method is able to realize more accurate localization of ship objects and the obtained bounding box contains less background, which is quite beneficial for the identification task. Compared with the SKG-Net, our proposed method shows a slight increase in

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TABLE VII
COMPARISON WITH DIFFERENT CNN-BASED METHODS ON THE LSSDD

Method	Anchor-free	Recall	Precision	F1	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.8}$	Time(ms)
R^2CNN [38]	×	0.7660	0.7009	0.7320	0.7520	0.3180	0.1720	117.2
Oriented R-CNN [39]	×	0.7404	0.7884	0.7636	0.6729	0.2369	0.1301	86.9
Rotated ATSS [40]	×	0.7772	0.7485	0.7626	0.7522	0.3602	0.2073	96.5
RoI Transformer [41]	×	0.7949	0.7861	0.7904	0.6902	0.2749	0.1414	106.5
S^2A-Net [42]	×	0.7692	0.7754	0.7723	0.7617	0.3121	0.1468	89.7
Oriented RepPoints [43]	✓	0.6266	0.5291	0.5737	0.5433	0.1828	0.0864	89.3
Oriented FCOS [44]	✓	0.7067	0.6923	0.6994	0.6245	0.2373	0.1025	74.9
Oriented CenterNet [22]	✓	0.7163	0.7376	0.7268	0.7410	0.3397	0.1893	**72.1**
SKG-Net [45]	✓	0.7612	0.7504	0.7558	0.7895	0.3650	0.2441	75.5
SRT-Net	✓	**0.8141**	**0.8038**	**0.8089**	**0.8271**	**0.4370**	**0.2786**	79.6

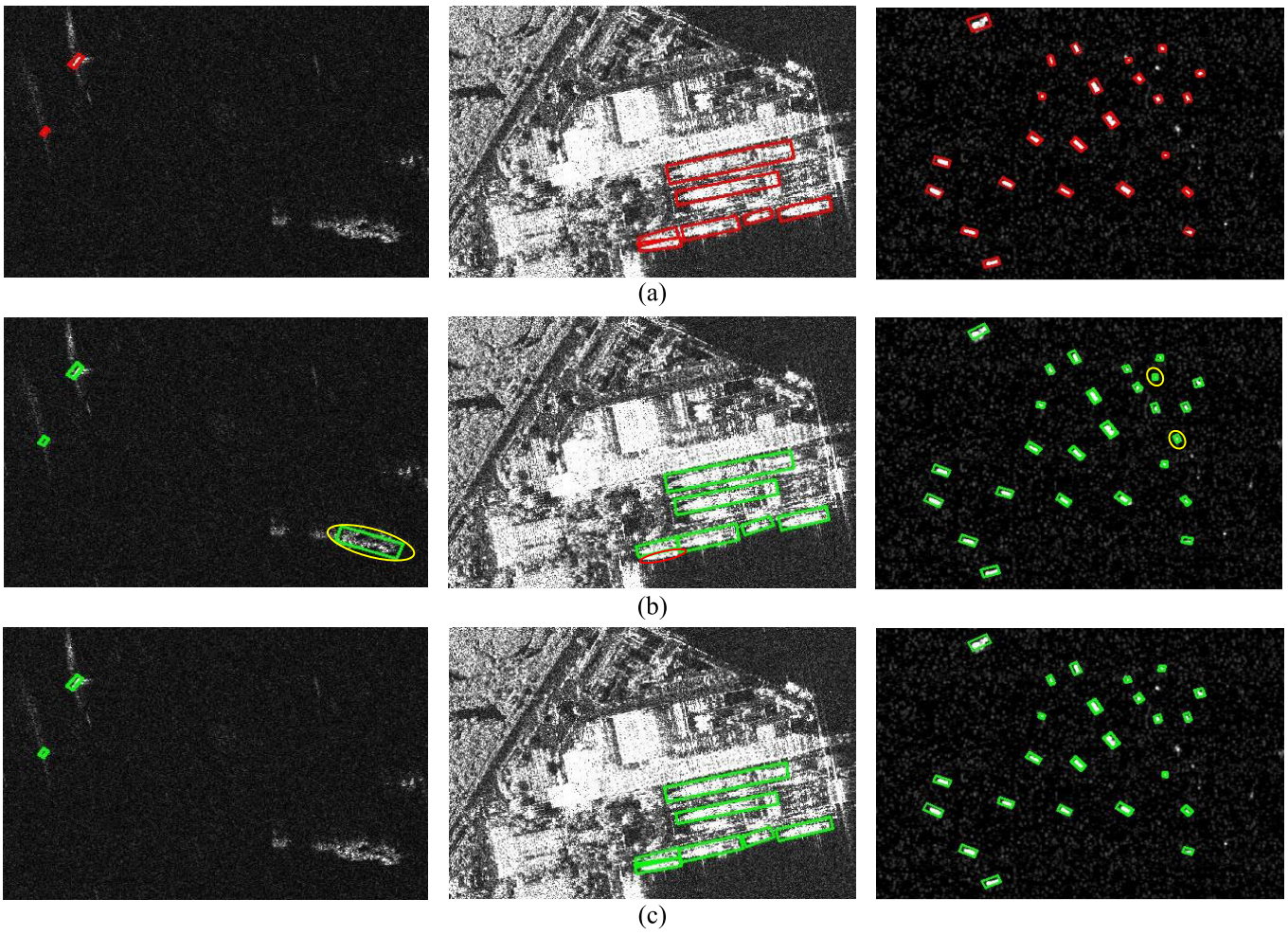
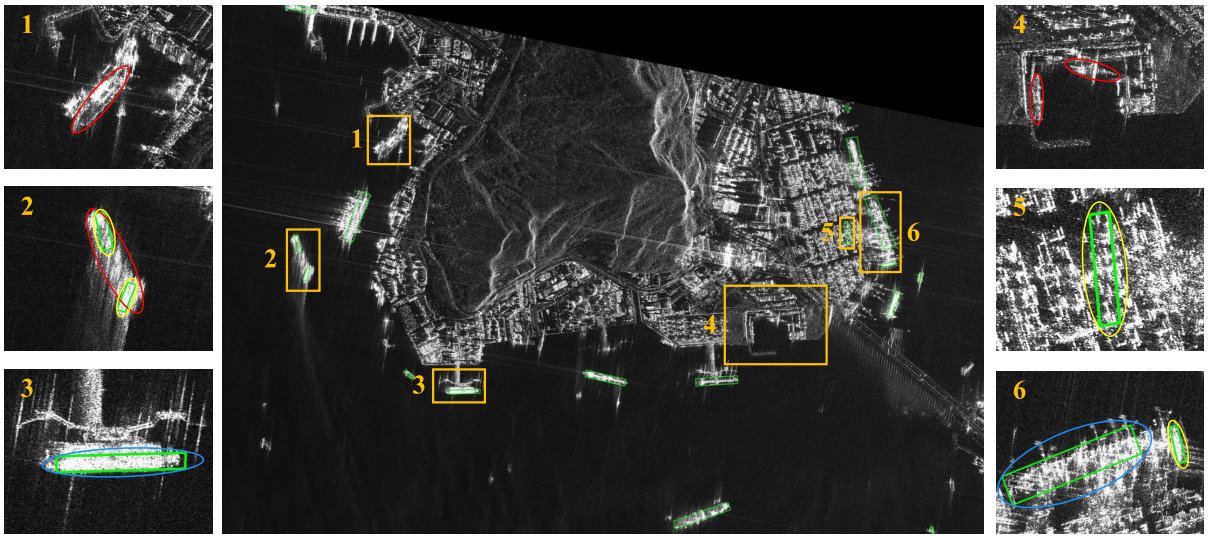


Fig. 20. Detection results on the SSDD+. Ground truths are marked in red boxes; detection results are marked in green boxes; missing ships are marked in red circles and false alarms are marked in yellow circles. (a) Ground truths. (b) Results of SKG-Net. (c) Results of our proposed method.

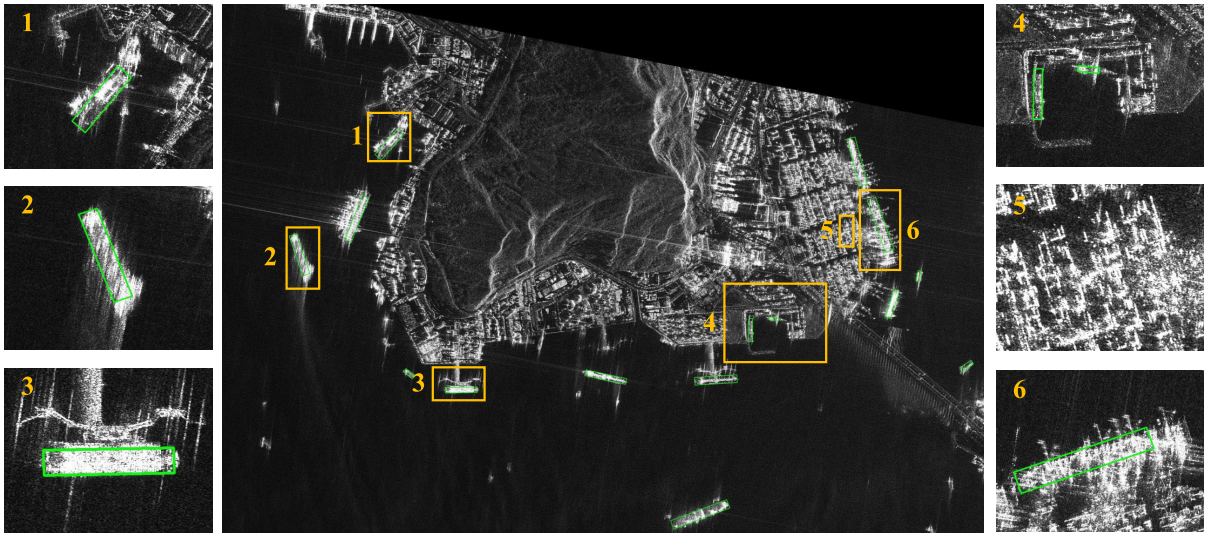
inference time but significant improvements in performance, which indicates that it is a better tradeoff between speed and accuracy.

To demonstrate the model's generalization and robustness, we conduct additional experiments on the public SSDD+ dataset and the results is presented in Table VIII. As shown in Table VIII, SRT-Net also achieves outstanding performance on this multiresolution SAR dataset, demonstrating its strong generalization ability. Fig. 20 shows some detection examples. Compared with SKG-Net, which generates some false alarms for objects that resemble ships, our proposed method avoids such errors by utilizing the global context from the constructed topology structure. Moreover, SKG-Net fails to detect some near-shore targets in complex scenes, while our proposed

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(a)



(b)

Fig. 21. Detection results in large-scale SAR images. Detection results are marked in green boxes; missing ships are marked in red circles; false alarms are marked in yellow circles and objects with low localization accuracy are marked in blue circles. (a) Results of SKG-Net. (b) Results of our proposed method.

TABLE VIII
COMPARISON WITH DIFFERENT CNN-BASED METHODS ON THE SSDD+

Method	Recall	Precision	F1	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.8}$
R^2CNN	0.9383	0.9472	0.9427	0.9288	0.3536	0.2045
Oriented R-CNN	0.9178	0.9195	0.9186	0.9260	0.3160	0.2136
Rotated ATSS	0.9084	0.9118	0.9101	0.9049	0.2459	0.1408
RoI Transformer	0.9028	0.8895	0.8961	0.9126	0.2873	0.1482
S^2A-Net	0.9103	0.9171	0.9077	0.9055	0.2529	0.1228
Oriented RepPoints	0.8411	0.8318	0.8364	0.8425	0.1934	0.1071
Oriented FCOS	0.8542	0.8978	0.8755	0.8801	0.2473	0.1318
Oriented CenterNet	0.9252	0.9287	0.9270	0.9151	0.3277	0.1781
SKG-Net	0.9402	0.9563	0.9482	0.9328	0.3751	0.2284
SRT-Net	**0.9514**	**0.9586**	**0.9569**	**0.9466**	**0.3927**	**0.2531**

method accurately identifies tightly arranged ships. These results suggest that SRT-Net can effectively emphasize the ships and enhance the detection performance.

For further verifying the effectiveness and practical applicability of SRT-Net, we perform comparison experiments on large-scale images and analyze the detection results. As the

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performance of SKG-Net is the second best of the above comparison experiments, we choose our proposed method and SKG-Net for experiments to analyze the detection performance on large-scale SAR images. In Fig. 21(a), some artificial facilities are misidentified as ship objects due to their texture highly similar to ships, and SKG-Net is hard to deal with ship objects under strong interference, resulting in missed detection or inaccurate localization. Compared with SKG-Net, our proposed method enhances the saliency of ships under strong interference, achieving the effects of reducing missing ships and accurate localization. In addition, our proposed method has fewer false alarms in the inshore regions, which shows the stronger anti-interference capability. Overall, our proposed method shows great advantages in detection performance and strong anti-interference capability in complex scenes.

V. Conclusion

In this article, a novel anchor-free method based on the scattering region topology network called SRT-Net is proposed for oriented ship detection in large-scale SAR images. The core idea of SRT-Net is to leverage the SRT-Net to dynamically capture a comprehensive global context and exploit the geometric contour information of ships, thereby enhancing ship saliency and effectively reducing false alarms. To address the challenges posed by complex scenes, the designed SRTP module constructs a topological structure, allowing for the extraction of richer semantic information in a dynamic manner, which significantly reduces false alarms. Additionally, the SSE module is introduced to guide the detector in learning ship contour information and structural features, leading to improved localization accuracy. Furthermore, the SFA loss is introduced to increase the attention to small-scale ships that are easily overlooked to improve the robustness of multiscale ship detection. To demonstrate the superior performance of our method, a series of comparative experiments on the LSSDD dataset and SSDD+ dataset show the superiority of our method.

References

[1] Y. Guan et al., "Fishing vessel classification in SAR images using a novel deep learning model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–21, 2023, Art. no. 5215821, doi: 10.1109/TGRS.2023.3312766.

[2] C. Zhang et al., "Evaluation and improvement of generalization performance of SAR ship recognition algorithms," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9311–9326, 2022.

[3] Z. Sun et al., "An anchor-free detection method for ship targets in high-resolution SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7799–7816, 2021.

[4] X. You and W. Li, "A sea-land segmentation scheme based on statistical model of sea," in *Proc. 4th Int. Congr. Image Signal Process. (CISP)*, vol. 3. Oct. 2011, pp. 1155–1159.

[5] J. A. Karvonen, "Baltic sea ice SAR segmentation and classification using modified pulse-coupled neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 7, pp. 1566–1574, Jul. 2004.

[6] G. Goldstein, "False-alarm regulation in log-normal and Weibull clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-9, no. 1, pp. 84–92, Jan. 1973.

[7] J. Ai, X. Qi, W. Yu, Y. Deng, F. Liu, and L. Shi, "A new CFAR ship detection algorithm based on 2-D joint log-normal distribution in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 806–810, Oct. 2010.

[8] R. Mamgain, R. Jain, D. Deb, and D. Seshagiri, "Two level CFAR algorithm for multiple target detection," in *Proc. 3rd Int. Conf. Converg. Technol. (I2CT)*, Apr. 2018, pp. 1–4.

[9] B. Hou, X. Chen, and L. Jiao, "Multilayer CFAR detection of ship targets in very high resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 4, pp. 811–815, Apr. 2015.

[10] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[11] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 2015, pp. 1137–1149.

[12] K. Zhou et al., "Arbitrary-oriented ellipse detector for ship detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 7151–7162, 2023, doi: 10.1109/JSTARS.2023.3267240.

[13] C. Li and H. Lang, "An oriented SAR ship detector with mixed convolution channel attention module and geometric nonmaximum suppression," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8074–8084, 2022.

[14] Z. Sun, X. Leng, Y. Lei, B. Xiong, K. Ji, and G. Kuang, "BiFA-YOLO: A novel YOLO-based method for arbitrary-oriented ship detection in high-resolution SAR images," *Remote Sens.*, vol. 13, no. 21, p. 4209, Oct. 2021.

[15] J. Dai, Y. Li, K. He, and J. Sun, "R-FCN: Object detection via region-based fully convolutional networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 2016, pp. 379–387.

[16] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[17] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[18] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 21–37.

[19] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[20] L. Huang, Y. Yang, Y. Deng, and Y. Yu, "DenseBox: Unifying landmark localization with end to end object detection," 2015, *arXiv:1509.04874*.

[21] X. Zhou, J. Zhuo, and P. Krähenbühl, "Bottom-up object detection by grouping extreme and center points," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 850–859.

[22] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[23] J. Fu, X. Sun, Z. Wang, and K. Fu, "An anchor-free method based on feature balancing and refinement network for multiscale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1331–1344, Feb. 2021.

[24] T. Zhang et al., "Balance learning for ship detection from synthetic aperture radar remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 182, pp. 190–207, Dec. 2021.

[25] K. Jin, Y. Chen, B. Xu, J. Yin, X. Wang, and J. Yang, "A patch-to-pixel convolutional neural network for small ship detection with PolSAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6623–6638, Sep. 2020.

[26] J. Zhao, Z. Zhang, W. Yu, and T.-K. Truong, "A cascade coupled convolutional neural network guided visual attention method for ship detection from SAR images," *IEEE Access*, vol. 6, pp. 50693–50708, 2018.

[27] X. Ke, X. Zhang, T. Zhang, J. Shi, and S. Wei, "SAR ship detection based on Swin transformer and feature enhancement feature pyramid network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 2163–2166.

[28] J. Zhang, M. Xing, G.-C. Sun, and N. Li, "Oriented Gaussian function-based box boundary-aware vectors for oriented ship detection in multiresolution SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art. no. 5211015, doi: 10.1109/TGRS.2021.3095386.

[29] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[30] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[31] C. Harris and M. Stephens, "A combined corner and edge detector," in *Proc. Alvey Vis. Conf.*, vol. 15, 1988, pp. 10–5244.

[32] S. Lloyd, "Least squares quantization in PCM," *IEEE Trans. Inf. Theory*, vol. IT-28, no. 2, pp. 129–137, Mar. 1982.

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[33] Y. Li and A. Gupta, "Beyond grids: Learning graph representations for visual recognition," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 9225–9235.
[34] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.
[35] Y. Chen, Z. Zhang, Y. Cao, L. Wang, S. Lin, and H. Hu, "Reppoints v2: Verification meets regression for object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 5621–5631.
[36] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.
[37] A. Neubeck and L. Van Gool, "Efficient non-maximum suppression," in *Proc. 18th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2006, pp. 850–855.
[38] Y. Jiang et al., "R2 CNN: Rotational region CNN for arbitrarily-oriented scene text detection," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 3610–3615.
[39] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3500–3509.
[40] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9756–9765.
[41] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2844–2853.
[42] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," IEEE*IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, Art. no. 5602511, doi: 10.1109/TGRS.2021.3062048.
[43] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented RepPoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1819–1828.
[44] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.
[45] K. Fu, J. Fu, Z. Wang, and X. Sun, "Scattering-keypoint-guided network for oriented ship detection in high-resolution and large-scale SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11162–11178, 2021.



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